PolyCo Screening Comparative Life Cycle Assessment of Circular Polyester Fiber

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Executive Summary

To advance its mission to revolutionize the textile industry with sustainability solutions, PolyCo has developed an innovative circular fiber recycling process. PolyCo, in collaboration with the sustainability consulting firm, OnePointFive, has developed a screening comparative Life Cycle Assessment (LCA) model designed to quantify and compare the environmental impacts of Polyco's circular polyester production process against traditional virgin polyester and bottle-to-fiber (BTF) alternatives.

The LCA model aims to:

- Evaluate the baseline environmental impacts of Polyco's circular polyester production from feedstock to yarn;
- Compare Polyco's innovative recycling technology with traditional and alternative polyester production methods;
- Guide strategic decisions based on environmental impact findings to minimize environmental impact of the technology at scale

This analysis evaluates three scenarios: Virgin Polyethylene terephthalate (PET) fiber, Bottle-to-fiber recycled PET (rPET), and Polyco rPET fiber (circular polyester), focusing on interim Greenhouse gas (GHG) emission estimates, hotspots, and emissions reduction potential. Although not third-party verified, the study aligns with the ISO 14040:2006 standard and the Higg Materials Sustainability Index methodologies. The below table summarizes our key insights & findings from the study

Moreover, Polyco's approach capitalizes on otherwise non-recyclable textile waste, offering a scalable and sustainable alternative in the textile recycling sector. The study underscores the importance of renewable energy and emission reduction strategies in production partnerships, which can further decrease Polyco's environmental impact. Additionally, the process's minimal chemical usage and waste, with a recycling mechanism for spent chemicals, aligns with Polyco's vision of a circular economy.

PolyCo's process is currently produced on a pilot factory level, and the findings within this report will be used to inform decision making around production locations, suppliers, and manufacturing partners, as PolyCo scales its operations. The findings within this report will also be expanded upon in a full-fledged LCA in Q1 2024.

1. Glossary of Terms and Definitions

Acronyms

Acronym	Term
BTF	Bottle-to-fiber polyester
CO_2	Carbon Dioxide (emissions)
CO_2eq	Carbon Dioxide equivalent (emissions)
EF	Emissions Factor
EG	Ethylene Glycol
EOL	End of Life
GHG	Greenhouse Gas (common unit CO_2-equivalent or CO_2eq)
ISO	International Standards Organization
KOH	Potassium Hydroxide
LCA	Life Cycle Assessment
PET	Polyethylene terephthalate (most common thermoplastic resin of polyester family)
rPET	Recycled polyethylene terephthalate (PET)

Definitions

Term	Definition
Avoided Emissions	A metric employed to express "the difference between GHG emissions that occur or will occur and GHG emissions that would have occurred without the solution," (WBCSD, 2023).
Boundary	Defines the limits of the system being assessed, noting what's included and excluded from the analysis.
Closed-loop source (recycling)	Manufacturing process that leverages the recycling and reuse of post-consumer products to supply the material used to create a new version of the same product (BEIS, 2023)
Comparative Life Cycle Assessment	Method used to evaluate and compare the environmental impacts of different products or systems over their entire life cycle. It helps identify the more environmentally preferable option among alternatives for informed decision-making.
Cradle-to-gate	Life cycle assessment from raw material extraction (cradle) to the factory gate intermediate product, defined in this study as cradle-to-yarn (see below)
Cradle-to-pellet	Life cycle assessment from raw material extraction (cradle) to an intermediate product, in this study, virgin PET / recycled rPET pellets pre-spinning.
Cradle-to-yarn	Life cycle assessment from raw material extraction (cradle) to an intermediate product, in this study, virgin PET / recycled rPET spun yarn.
End of Life (EOL)	Refers to the life cycle stage of a product or service when it is no longer being used by a consumer, e.g. recycling or landfill processes after product use.
Functional Unit	A quantified description of the service that a product provides, used as a reference unit in the LCA study to ensure comparability of results.
Functional Use	Stage of customer utilization of a specific product where the manufactured good performs its purpose (i.e.. Polyester life cycle phase where it is used as clothing)
Goal	Refers to the specific purpose and intended application of the LCA study, guiding the scope and depth of the analysis.
Life Cycle Assessment	Process for assessing the environmental impacts associated with all stages of a product's life, from raw material extraction to disposal. It aims to understand and reduce a product's total environmental impact.
Open-loop source (recycling)	Manufacturing process where materials are recycled and reused into post-consumer products that are different than the original (pre-recycling) (BEIS, 2023)
Scope	Delineates the extent and boundaries of the study, including the product system to be studied, the life cycle stages to be considered, and the types of impacts to be assessed.

2. Introduction & Disclaimer

PolyCo is in the process of developing a production method for circular polyester with significant decarbonization potential. To evaluate the potential impact of their technology at-scale, PolyCo prepared a screening comparative LCA model. This initial screening LCA model aims to identify hotspots and compare PolyCo's fiber production from feedstock to yarn at-scale, using assumptions, against virgin polyester and bottle-to-fiber alternatives.

The screening model was developed over a six-week period in collaboration with OnePointFive, a sustainability consulting firm headquartered in New York. The model intends to:

- Quantify baseline environmental impacts of PolyCo's process for producing circular polyester and alternatives;
- Compare PolyCo's target technology against main alternatives of virgin polyester and BTF polyester; and
- Inform strategic decision-making with environmental impact findings, including selecting production locations, transportation methods, and energy generation sources.

The model evaluates three key scenarios for comparison, namely, Virgin PET fiber, BTF rPET (open-loop alternative), and PolyCo rPET fiber (closed-loop recycling innovation). The purpose of the model is to identify interim GHG emission estimates, hotspots, comparisons amongst key technologies, and potential for lower emissions generated through use of PolyCo rPET fiber.

The goals of the study are to identify opportunities for environmental impact reduction within PolyCo's control, and to obtain baseline environmental awareness and proof on circular polyester's viability. **This study follows the ISO 14040:2006 standard & Higg Materials Sustainability Index, but has not been third-party verified or validated.**

Hence this study should be used for internal purposes only, and for sharing product data with business partners. It should **not be used for public communications and/or claims,** nor presented as the output of a thorough investigation into all aspects of the product's environmental impact.

3. Approach & Methodology

Screening Comparative Life Cycle Assessment Model

This LCA model estimates and compares three distinct technologies for producing polyester fiber, distinguishing between:

1. Production of Virgin PET pellets and yarn
2. Alternative existing technology of recycled rPET bottle-to-fiber (open-loop recycling)
3. Alternative innovative technology of PolyCo rPET circular fiber (closed-loop recycling)

The methods used to develop the model generally follow the process LCA standards developed and maintained by the International Standards Organization (ISO), specifically ISO 14040:2006 and ISO 14044:2006, and the Higg Materials Sustainability Index and Product Module. Following the ISO 14040 guidelines, the model includes key components of: goal and scope definition, life cycle inventory analysis, impact analysis and interpretation.

Important Note: This Screening Comparative LCA Model is an initial assessment intended for internal audiences and key stakeholders. The model has not been third-party validated or verified. External or public claims should not be made from the model outputs & results without GHG Accounting expert guidance, audit and review.

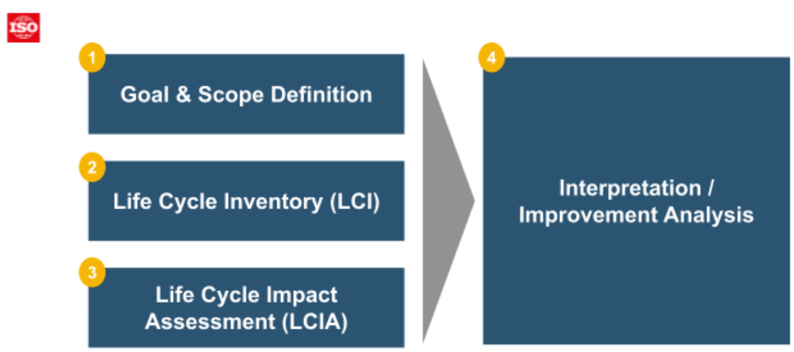


Figure 3.1: ISO 14040:2006 defines four steps for conducting an LCA (ISO 14040:2006, Lamperti Tornaghi et al, 2018)

Aligned with ISO standards guidance, the collection of primary data was prioritized, before the use of secondary, proxy or estimate data. When non-primary data was used, the sources, calculations (as relevant), and assumptions were clearly noted. As PolyCo's technology is not yet at commercial scale as of December 2023, a significant number of assumptions were made to complete this study. Both Virgin PET and recycled rPET BTF life cycle assessments relied on secondary activity data and emission factors. See the Appendix for more information.

Emissions factors (EFs), when utilized, were sourced from Ecoinvent database v3.9.1. All EFs sourced from Ecoinvent follow the IPCC 2021 methodology. The EFs used were aligned to the sourcing locations for materials, when this information and EF were available. When data gaps pertaining to material sourcing geographies were identified, EFs for Global (GLO) or Rest of World (RoW) markets were used. More information regarding the specific EFs sourced can be found in the “EF” sheet in the *PolyCo Screening LCA* (Excel file).

Scope and Boundaries

According to ISO 14040 standards, the scope of an LCA study defines what aspects of the product, process, or service will be analyzed. The scope includes the goals, objectives, and system boundaries which structure the unit(s) of comparison within the study. Furthermore, LCA guidance from the Higg Index calls for: unambiguous description of classification and characterization model applied, the system boundary, and the allocation or system expansion applied in the model. The below figure summarizes the scope and boundaries of the Screening Comparative LCA model, aligned to the ISO 14040 and Higg Index guidance.

Goal	Build a screening Comparative LCA model to evaluate emissions & hotspots of Virgin fiber, PolyCo fiber, and Bottle-to-fiber, to present to investors	
Objectives	Quantifying the environmental impact of each scenario to compare outcomes; identifying areas for future study to include in follow-up LCA study	
Functional unit	1kg of PolyCo circular polyester yarn through its entire lifecycle	
Scope	**Includes** • **Transportation**, from raw materials to factory only • **Raw Materials**, feedstock • **Manufacturing**, polymerization & depolymerization, yarn spinning) • **End of Life**, for yarn/fiber	**Excludes** • **Transportation**, post factory • **Distribution**, beyond transportation of raw materials • **Customer use** phase
System Boundaries	From cradle-to-gate: feedstock to yarn production, including end of life treatment	

Figure 3.2: Overview of the goal, objectives, unit of measurement, scope and system boundaries for this study

Three process flows were developed to map out the cradle-to-gate steps for each technology assessed. Inputs and outputs into each process step, including energy usage and waste production & handling, were defined in the diagrams. The diagrams emphasize comparability across the three technologies given matching scopes and system boundaries, and were then used as the basis to develop LCA models for each technology.

The PolyCo process flow is included below for context. See the Appendix for the process flows depicting the Virgin PET and recycled rPET BTF process flow diagrams.

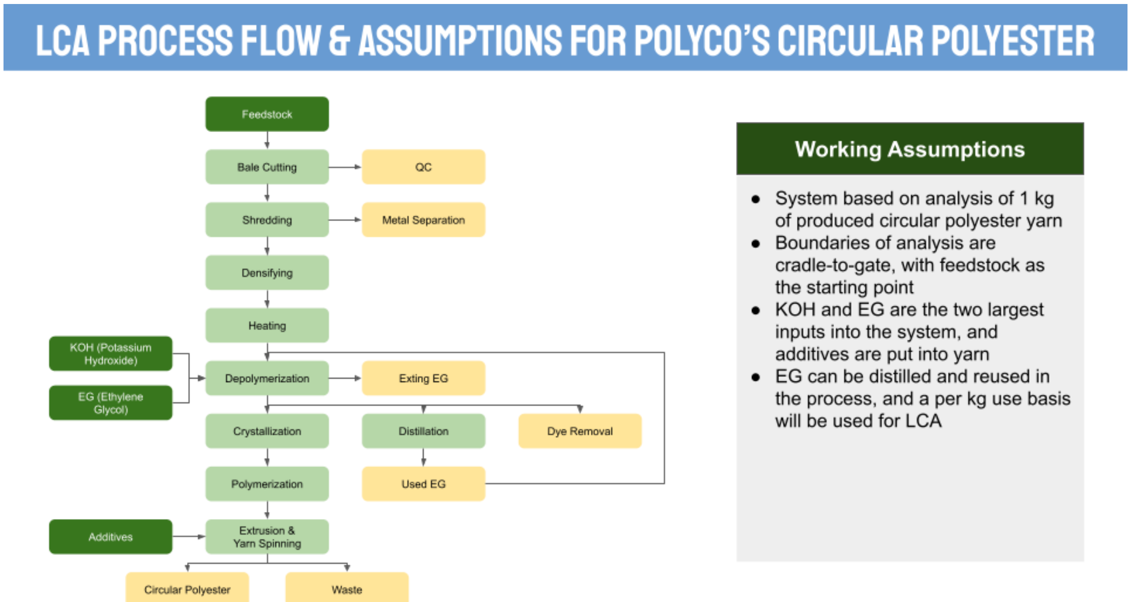


Figure 3.3: Life cycle inventory process flow diagram of the target technology, PolyCo circular fiber. Inputs in dark green, outputs in yellow, and processing steps in light green

Comparison Basis

The PolyCo yarn has a longer life cycle compared to the virgin and BTF alternatives, due to their inability to replicate the same recycling capacity. Therefore, adjustments to the Split of Functional uses were made to ensure comparability of emissions outputs across the three technologies. The following clarifies the Raw Materials and EOL life cycle processes for the three technologies, expressing how they all require the use of raw virgin polyester as a critical component:

Virgin PET
- Fossil raw materials are extracted and pre-processed (cradle-to-pellet) with eventual EOL treatment (i.e. incineration)

Bottle-to-fiber rPET
- Virgin PET production is a precursor (input) for the BTF rPET processes (i.e., open-loop recycling),
- Bottle PET is also a feedstock (input) to produce spun yarn or fiber (output), which affects the comparison basis to PolyCo's closed-loop source recycling technology

PolyCo rPET
- Virgin PET production is a precursor (input) for the use of PolyCo's recycling technology (i.e., closed-loop recycling), which is accounted for in modeled outputs.

- PolyCo's recycling technology diverts PET fabric waste from incineration, and recycles it up to a proven 8 times

While virgin polyester is a critical component across the three technologies, the inclusion of this virgin material across the three technologies is distinct. Virgin PET production assumes a 1 functional *use* because the product is disposed of (incinerated), as opposed to recycled after its initial consumer use. The BTF process and PolyCo innovation have higher recycling capacities, compared to the production of virgin polyester. For example, PolyCo's technology (closed-loop source recycling) has been proven to maintain virgin quality for eight (8) recycles for every one functional unit (1 kg of material) of virgin PET used; the split of functional uses therefore shows that for the PolyCo fiber, there is 1 functional use as virgin polyester and 8 functional uses as circular polyester. PolyCo therefore delivers nine (9) functional uses to the market per 1 kg PET.

Adjustments were made to scale virgin PET and BTF to ensure comparability across the technologies on a 1:1 basis. Looking at the table below, the PolyCo fiber has 1 life cycle because it is the basis of the defined functional unit: 1 kg of PolyCo circular polyester yarn throughout its entire lifecycle. However, the split of functional uses clarifies how adjustments were made to accommodate for the *difference in functional uses* across the technologies: while the first product created in the PolyCo fiber scenario is made from virgin polyester, the succeeding 8 products (functional uses) created are made from circular polyester post-recycling. Adjustments to calculations therefore account for the split of functional uses for all technologies, given variable recycling capacities that impact raw materials and EOL processes and emissions outputs.

Technology	PolyCo	Virgin	Bottle-to-Fiber
Life Cycles	**1 - Basis of LCA Functional Unit**	9	4.5
Split of Functional Uses	1 as Virgin Polyester, and 8 as Circular Polyester	9 as Virgin Polyester	4.5 as Bottle, and 4.5 as Fiber
Functional Uses (products)	9	9	9

As specified by the table, the alternative to PolyCo's functional unit is to produce 9 kg of virgin PET yarn (cradle-to-yarn). For BTF recycling (open-loop source recycling), the alternative to PolyCo yarn is to produce 4.5 kg of yarn using the BTF technology from 4.5kg of Bottles. The BTF recycling can only be performed once for the same unit of PET, as it relies exclusively on *bottled* PET as feedstock, and then produces *yarn* as output.[1] To make BTF recycling

[1] Bottle-to-fiber technology cannot be used to recycle the ~63 million tonnes of polyester fiber produced annually. Source: Polyester fiber production globally 1975-2022 - Statista https://www.statista.com › ... › Chemical Industry

comparable with PolyCo on a *cradle-to-pellet* basis, the BTF LCA screening results are split (scaled) to reflect 4.5 functional uses of virgin PET pellets for bottles and 4.5 functional uses of rPET pellets for yarn. The bottle-to-fiber material thus encapsulates 4.5 life cycles to reflect a 1:1 comparison with the specified functional unit of PolyCo's circular recycling.

Key Limitations

Detailed assumptions are summarized in the Appendix, and within the emissions model itself.

Summary of Key Limitations
Use of proxies and projections to emulate the emissions of a full-scale factory based on a smaller pilot-level factory
Academic studies and emissions factors used to replicate the comparison of the target technology to alternatives, virgin polyester and bottle-to-fiber recycled polyester
Transportation analysis assumes estimates of likely distances between prospective factories, and location of an initial factory located within the EU, with exact location to be informed by the full-fledged model outputs at a future date

Use of Proxies & Projections

While PolyCo has studied its process and has been able to produce certified results showing an upkeep of virgin polyester quality through eight (8) recycles, it has yet to scale the technology to a production level larger than its current pilot factory size. The LCA model was designed to be representative of circular polyester's impact when produced at scale in a future European factory, capable of processing 250kt of circular polyester pellets annually. The model utilizes primary estimates of the machinery and energy usage that would be required to produce such a quantity.

Academic Studies & Emissions Factors

The comparison made between the two non-target technologies is largely based on secondary data and research. While PolyCo has detailed insights into its own process's inputs, outputs, machinery usage, electricity, and projections for expected transportation methods, the same granularity of data was not available for virgin polyester nor BTF manufacturing processes.

In these instances, the LCA relies on emissions factors and academic research:

- In the model of virgin polyester production, Ecoinvent 3.9.1 was utilized for the process and simplified to have inputs of purified terephthalic acid (PTA) and ethylene glycol (EG). The process was assumed to use renewable energy, to maintain a consistent comparison across the three technologies.

- In modeling BTF polyester, emission factors from a 2011 study at Utrecht University were used to model both the PET-to-bottle process and the bottle-to-pellet process

Transportation Analysis

The lack of physical infrastructure currently developed by PolyCo acts as a practical limit to the analysis of transportation of material. While transportation represented a very small portion (<2%) of emissions from the PolyCo production process, it was modeled (from PolyCo guidance) anticipating that the feedstock would travel ~50km on truck, which may be subject to change depending on the agreements that follow PolyCo's planned expansion.

4. Key Results & Insights

The study categorizes and outlines the key results & insights of the study as summarized in the table below.

	Key Results & Insights Areas
Technological Achievements of PolyCo Circular Polyester	• Cradle-to-Pellet + EOL Comparison with Virgin Fiber & Bottle-To-Fiber • PolyCo Circular Polyester Process & Emissions
Recycling Importance for PolyCo Circular Polyester	• Cradle-To-Yarn + EOL Comparison with Virgin Fiber • PolyCo Sensitivity Analysis for Number of Recycles

Technological Achievements of PolyCo Circular Polyester

Cradle-to-Pellet + EOL Comparison with Virgin Fiber & Bottle-To-Fiber

The following section evaluates the results of the Screening Comparative LCA emissions model assessment for the three defined scenarios. The model analyzed the baseline emissions for the virgin polyester production process, the emissions of the alternative BTF process, and the emissions for PolyCo's innovative circular polyester production process.

In the initial comparison of the three scenarios from cradle-to-pellet with EOL, the benefits of PolyCo's circular polyester are displayed most clearly. This initial analysis concludes that, compared to virgin polyester production, **PolyCo pellets are projected to generate 33.629 kg less CO_2eq emissions than virgin pellets over their lifecycle** (assuming 9 functional uses and 1 EOL process). This equates to 76.2% potential avoided GHG emissions relative to the same boundary measuring the production of 9 kg virgin Polyester pellets (refer to the Comparison Basis above).

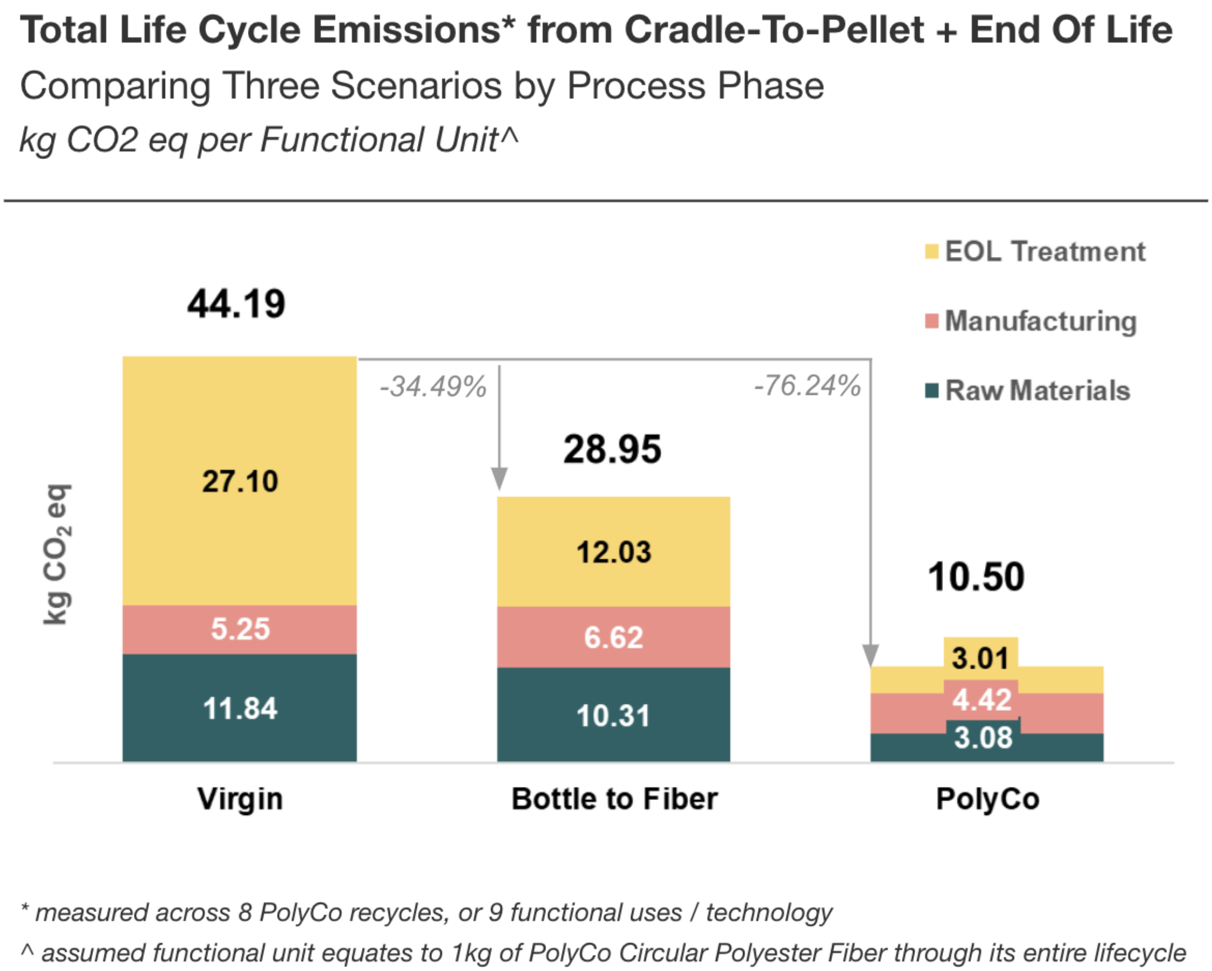


Figure 4.1: Cumulative GHG emissions of nine functional uses (refer to table in "Comparison Basis")

PolyCo's production is not a direct alternative to bottle-to-fiber rPET due to feedstock utilization, given that BTF reflects open-loop source recycling that utilizes bottles and not fabric. However, producing circular polyester versus BTF rPET could potentially generate 18.45kg less CO_2eq per 1kg of circular polyester pellets over the pellets' lifetime (9 functional uses).

This analysis was also produced to demonstrate the relationship and emissions across a single functional use basis, which is detailed below in Figure 4.2.

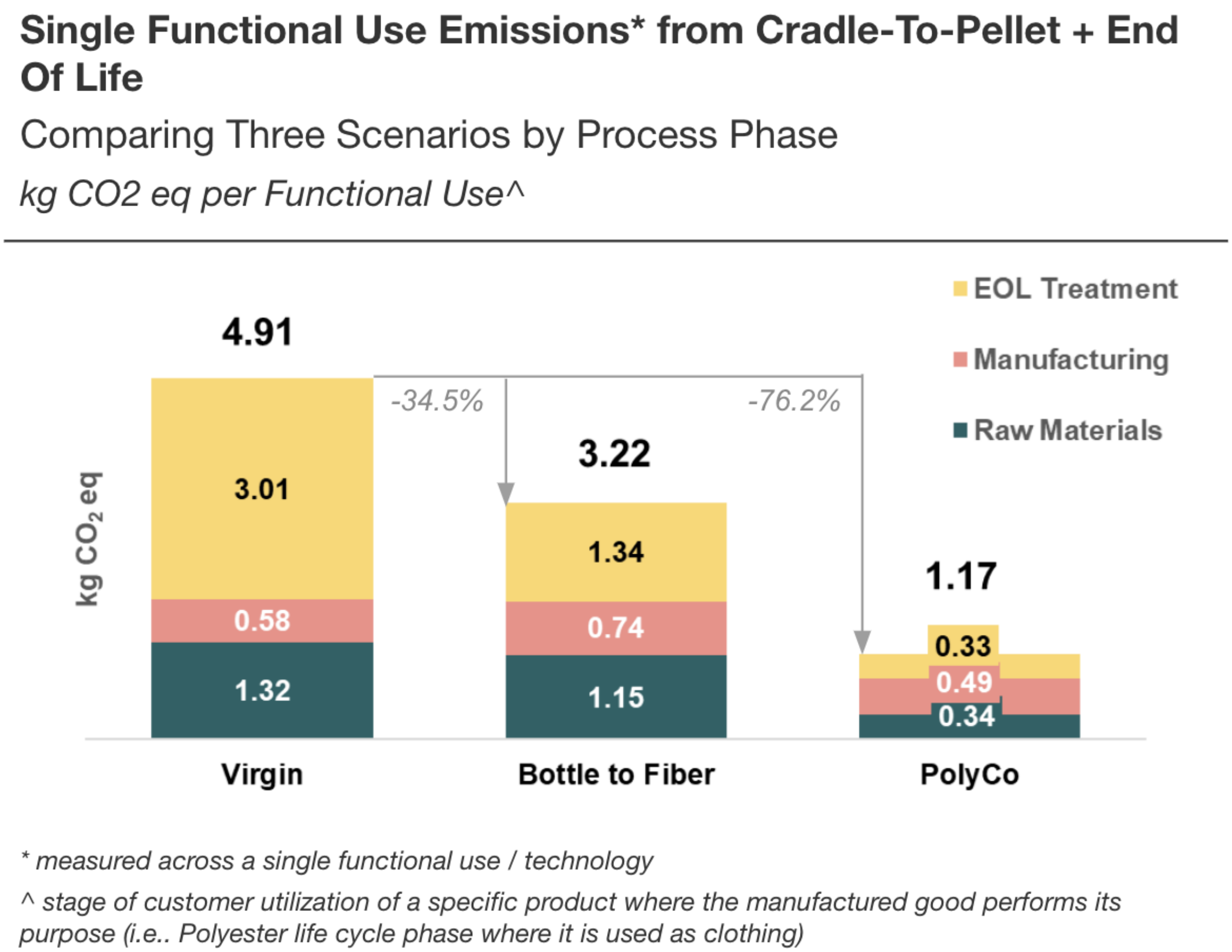


Figure 4.2. ***Normalized comparison per single functional use****. Multiplying each of the single use figures in this chart by the number of functional uses (i.e., 9) yields the cumulative GHG emission figures in Figure 4.1*

While PolyCo's most direct comparison is to be made with virgin polyester, it demonstrates clear benefits in terms of raw materials and EOL treatment over both virgin and bottle-to-fiber production. For the comparisons made in this analysis, a common EOL treatment of incineration was used, as PolyCo will be utilizing fabrics that were otherwise labeled for incineration in its production facilities. Since PolyCo's fabric would be incinerated only once following its 9 functional uses, this represents a significant advantage over virgin polyester, which would incinerate an end product 9 times. BTF is recyclable only once before significant drops in quality occur, so emissions from EOL treatment are considerably lower compared to virgin polyester due to fewer assumed incinerations.

Differences in emissions attributed to raw materials are also due to the respective products' recyclability – PolyCo can produce 9 functional uses from one initial batch of polyester raw materials, whereas virgin polyester repeats the raw material collection process each time. BTF requires 4.5 batches of raw materials to model the equivalent 9 functional uses of 1 kg of PolyCo's circular polyester, and its raw material requirements are understandably high, given the processing of bottles that precedes pelletization.

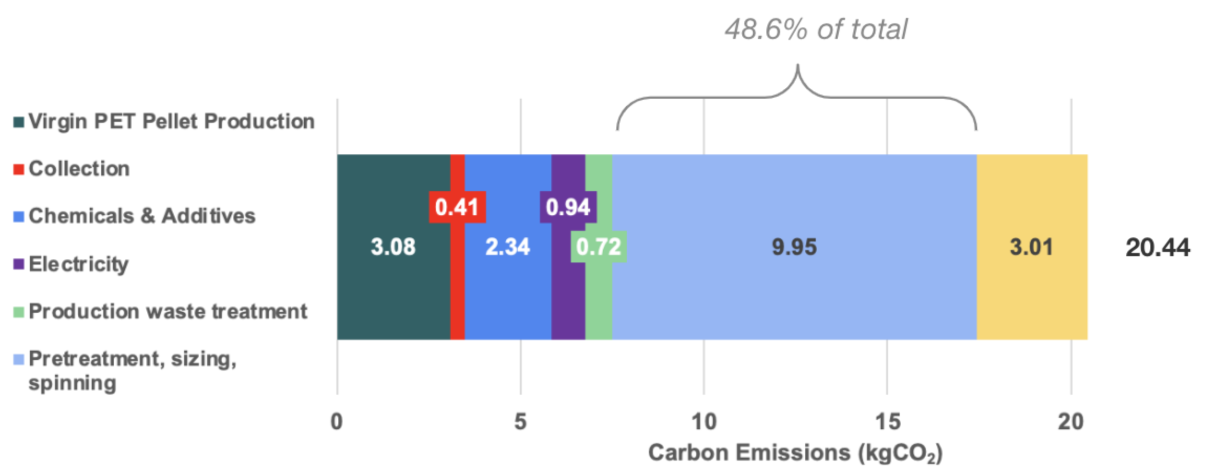
PolyCo Circular Polyester Process & Emissions

A closer analysis of the circular polyester process illustrates the potential emission hotspots of the PolyCo feedstock-to-yarn process. This analysis expanded the view of the technology to include yarn spinning (i.e. cradle-to-yarn). Yarn spinning emissions were assumed to be similar across the three scenarios, with minor differences from spinning recycled vs. virgin polyester. Figure 4.3 explores the emissions through processing, from cradle-to-yarn plus EOL, of the functional unit over 9 functional uses.

Total Life Cycle Emissions* from Cradle-To-Yarn + End Of Life

Focus on PolyCo Circular Polyester Production Process

*kg CO2 eq per Functional Unit^*



** measured across 8 PolyCo recycles, or 9 functional uses / technology*

^ assumed functional unit equates to 1kg of PolyCo Circular Polyester Fiber through its entire lifecycle

Figure 4.3: Overview of the goal, objectives, unit of measurement, scope and system boundaries for this study

This analysis determined that, assuming a lifecycle involving 8 recycles, the largest emissions hotspot in PolyCo's process would be derived from pre-treatment, sizing, and spinning, given that an average global energy grid mix was used to calculate emissions from spinning operations. Following spinning, the initial virgin PET raw material constitutes the second-largest carbon impact, followed by EOL treatment (incineration) and the chemicals included in the production process. As PolyCo identifies and selects its future production partners, it will be crucial to form yarn spinning agreements with suppliers that prioritize renewable energy usage and low-emission operations. If produced using renewables, the emission factor used for these suppliers' operations would be less emissions-intensive than the average global energy grid mix, which could result in reducing up to the current 9.95kg CO_2eq per functional unit attributed to pre-treatment, sizing, and spinning in the PolyCo production process.

Importance of PolyCo Circular Polyester Recycling & Yarn Spinning

Cradle-to-yarn + EOL Comparison with Virgin Fiber

Spinning, the final step from pellet to yarn, can be a large component of the cradle-to-yarn plus EOL emissions footprint for each of the three scenarios analyzed. It was assumed that all pellet production pathways rely on the same energy-source (e.g., oil or solar PV) for spinning processes for the sake of fair comparison. Spinning recycled rPET versus virgin rPET is slightly more energy-intensive according to sourced proxies, however, this does not materially impact the comparison.

On the basis of a thorough comparison with virgin polyester, which is PolyCo's primary alternative given that BTF utilizes different feedstock, the following figures were constructed to demonstrate the benefits of PolyCo's circular polyester as the fabric is recycled. Figure 4.4 demonstrates the main differences between virgin and circular polyester from cradle-to-yarn with EOL included, with the same 33.62 kg CO_2eq difference between production processes. Spinning, assumed to be using a consistent global energy mix across both technologies, is included in this analysis utilizing an EU fabric spinning proxy. BTF yarn was excluded from the analysis, as it only experiences 4.5 spinning cycles over the functional unit life cycle, which makes for an uneven comparison.

The fraction of the cradle-to-yarn GHG footprint of any PET spun yarn is highly dependent on the energy source used to spin the PET. If the energy is generated using fossil fuels, it may account for a sizable fraction of GHG emissions for a cradle-to-yarn plus EOL boundary. The key takeaway for PolyCO is to consider the potential impact of the energy source used by spinners as it forms future partnerships.

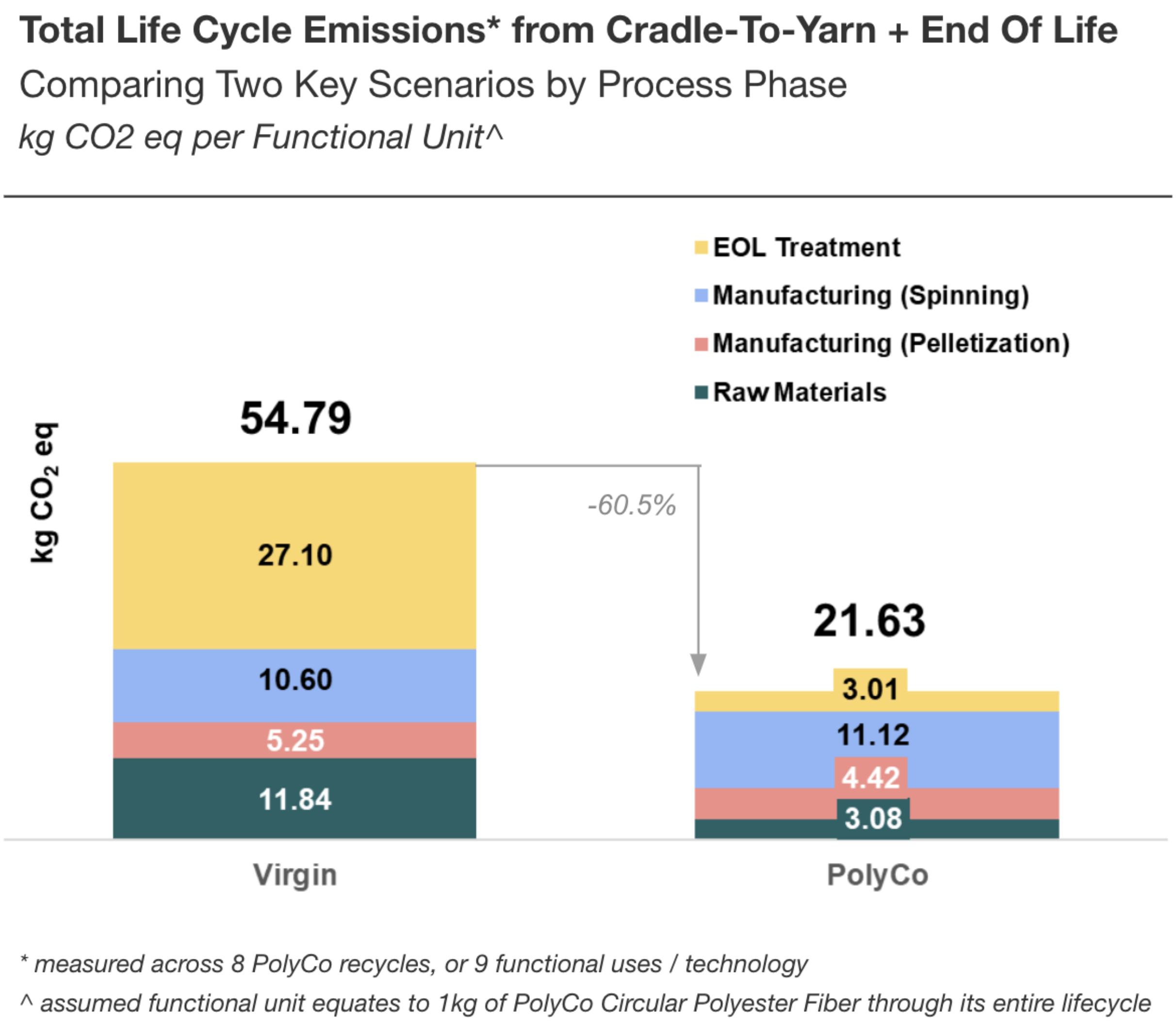


Figure 4.4: Overview of the goal, objectives, unit of measurement, scope and system boundaries for this study

PolyCo Sensitivity Analysis for Number of Recycles

Finally, many of the model-driven avoided emissions from PolyCo's circular polyester (as compared to using virgin polyester) are accrued over time as the circular polyester undergoes additional recycling processes and replaces the need to produce new virgin polyester. Figure 4.5 below was constructed to explore this relationship from cradle-to-gate across up to 10 recycling processes being completed. This sensitivity analysis illustrates the quantitative relationship of avoided emissions of PolyCo circular polyester compared to virgin polyester from cradle-to-yarn (+EOL) for every consecutive recycle.

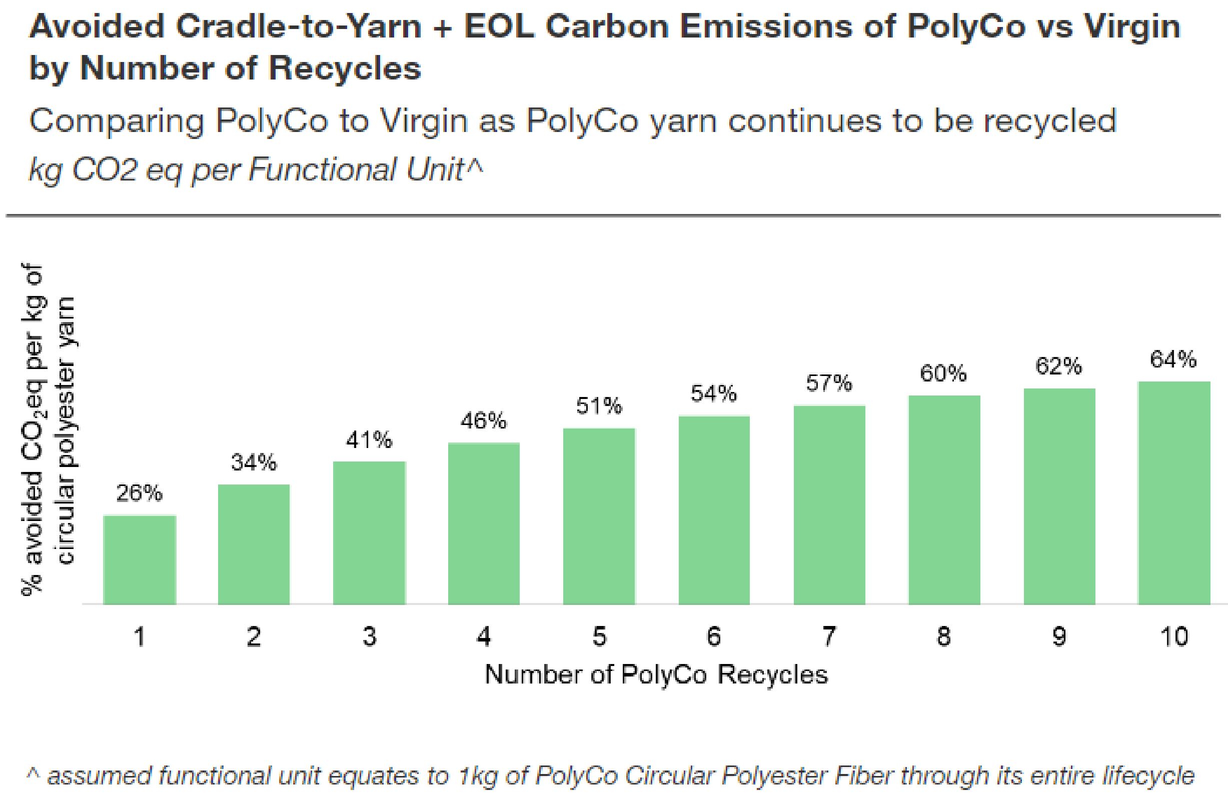


Figure 4.5: Cradle-to-yarn plus EOL analysis of avoided emissions as PolyCo circular polyester is recycled

While PolyCo saw no reduction in recyclability or polyester quality when testing its process in the lab over 8 recycles, the relationship present in this graph demonstrates that the circular polyester process can potentially avoid at least 26% of emissions compared to virgin polyester in its first recycling process, with this number increasing with further recycles. However, the potential avoided emissions begin to plateau to approximately 64% from 10 cycles onwards, exhibiting diminishing marginal returns. The potential avoided emissions are significant, as existing recycling methods focus on using feedstocks of bottles, rather than textiles, and any emissions avoided from the abated disposal of would-be-incinerated polyester represent large gains to be made in apparel, footwear, and plastic supply chains.

5. Conclusion & Other Considerations

Relying primarily on inputs from Ecoinvent 3.9.1, academic research, and primary inputs from the PolyCo manufacturing team, this screening comparative LCA has determined that PolyCo's circular polyester could potentially avoid at least 26% of cradle-to-yarn plus EOL emissions from virgin polyester production per one recycle. Furthermore, PolyCo's lab-proven 8 recycling processes can translate to potential avoided emissions of 60% from virgin polyester across 9 functional uses, or a total of 33.62 kg CO_2eq using a cradle-to-yarn plus EOL boundary. While PolyCo and BTF production processes utilize different inputs (fabric vs bottles), PolyCo's process projects to generate 18.45 kg less CO_2eq than BTF processing over 1 kg of circular polyester's lifecycle, a 63.7% reduction, when analyzed from cradle-to-pellet plus EOL.

Insights	Key Results
Avoided Emissions Compared to Virgin Polyester	Considering a *cradle-to-pellet* + *EOL boundary*, PolyCo's circular polyester could avoid 76.2% of the GHG emissions associated with virgin polyester, saving 33.629 kg CO_2eq per 1 kg of PolyCo circular polyester over its lifecycle.
PolyCo Compared to Bottle-To-Fiber	Considering a *cradle-to-pellet* + *EOL boundary*, PolyCo's process was projected to result in 18.39 kg fewer CO_2eq compared to BTF processing per 1 kg of circular polyester's lifecycle, a 63.7% reduction.
PolyCo Emissions Hotspots	Considering a *cradle-to-yarn* + *EOL boundary*, the largest emissions hotspot (48.6% of total) in PolyCo's process was determined to be pre-treatment, sizing, and spinning, given an assumption of average global energy grid mix for supplier partners.
Avoided Emissions for One (1) Recycle vs. Virgin Polyester	Considering a *cradle-to-yarn* + *EOL boundary*, PolyCo's circular polyester projects to avoid at least 26% of emissions from virgin polyester production with only one recycle.
Avoided Emissions for Total (8) Recycles vs. Virgin Polyester	Considering a *cradle-to-yarn* + *EOL boundary*, PolyCo's lab-proven 8 recycling processes can translate to 60% avoided emissions from virgin polyester.

Other proven benefits to PolyCo processes include repeatable virgin quality for the entirety of the circular polyester's lifecycle, known already to last at least 8 recycles. Repeatedly high quality differentiates the PolyCo circular polyester from BTF, which shows deteriorating quality over consecutive recycles.

Additionally, PolyCo's raw material feedstock would otherwise be unusable, as there are limited commercial-level technology solutions for utilizing or upcycling discarded polyester fabric with low emissions intensity. PolyCo's technology has the potential to close the loop of fabric waste and repeatable upcycling fabric back to assumed virgin polyester quality, giving it a significant advantage over the open loop, limited recycling of BTF, and high emissions intensity of virgin polyester production.

Finally, PolyCo's process utilizes few chemical inputs and produces minimal chemical waste given the majority is recycled back into the production process. Currently, the process utilizes ethylene glycol (EG), potassium hydroxide (KOH), as key process-driving inputs, and low quantities (~5% end weight) of additives such as titanium oxide in the coloration process. Spent chemicals that are not included in the end product, such as EG used in processing capacities only, are recycled, and are usable in later manufacturing contexts.

6. Recommendations & Future Considerations (for PolyCo Team Only)

This study set out to explore the differences in life cycle emissions between virgin polyester, bottle-to-fiber polyester, and PolyCo's circular polyester at a screening level, giving investors and LCA experts insight into the outputs of the process as it is developed and verified in laboratory applications and tests.

The current analysis will be built upon in a full-fledged LCA beginning in Q1 2024 that will further explore the three processes in much greater detail and scrutiny.

This study has revealed significant emissions reductions enabled by the development of PolyCo's circular polyester compared to both virgin polyester and BTF polyester. The leadership team at PolyCo, stressing the uses of fibers that would otherwise be incinerated and sourcing renewable energy in production, combined with the ability to produce virgin-quality polyester in perpetuity, enable Polyco's circular polyester to project a 26% decrease from virgin polyester cradle-to-gate (+ EOL) emissions after 1 recycle and a 60% decrease in cradle-to-gate (+EOL) emissions over virgin polyester after the laboratory-proven 8 recycles. The technology also projects to emit 18.39 kg CO_2eq less than BTF processes over the life cycle of 1 kg circular polyester using a cradle-to-pellet (+ EOL) boundary.

In the future, a closer analysis should target cradle-to-grave emissions in demonstrating the benefits of PolyCo technology in practical applications. In future analyses, PolyCo could consider the following:

- analyzing production & distribution locations,
- evaluating the impact of different polyester disposal methods,
- addressing process waste treatment more closely,
- evaluating spinning partners & primary energy sources of associated processes, and
- identifying closer proxies or primary data on the virgin and bottle-to-fiber production processes

7. Appendix

Assumptions and Limitations

The screening LCA was built using a mix of primary data, Ecoinvent emissions factors, and activity data from peer-reviewed academic research, and is limited to an exploratory level of data quality. The model and resulting analysis has yet to be third-party verified, but has been refined over multiple iterations of emissions factor search to produce a replicable and actionable result for the PolyCo data team and prospective investors.

Key assumptions that affect the model's outputs in a material way include:

- The method of **EOL disposal** used for polyester influences cradle-to-yarn plus EOL emissions of all technologies. Varying disposal methods (incineration, open dump, open burning, and/or landfilling) would have carbon and non-carbon related effects on the life cycle impact of circular polyester when compared to alternative technologies. Incineration using an Ecoinvent emission factor for Switzerland was selected as the common EOL method for each, to maintain comparability with the avoided EOL of PolyCo's feedstock.

- The **production electricity** used for each technology, as all pellet production was assumed to be done using renewable energy. **Spinning**, specifically, was assumed to use a global energy mix, to account for the historically large impact that spinning has had for polyester yarn spinning, and which it could have on the PolyCo production process in the absence of a renewable energy spinning source used by a production partner. Several sources were researched, including secondary emission factors from Ecoinvent and both activity data and emission factors from peer-reviewed papers were studied to determine the activity data and emission factors modeled for spinning. A recent study in the *Clean Technologies and Recycling* via AIMS Press[2] highlights the variation in spinning energy and GHG emission-intensity of spinning processes. The model can be updated to rely on varying energy sources to develop a screening sensitivity that shows the impact of low-carbon energy sources for spinning on the cradle-to-yarn plus EOL boundary life cycle of PolyCo and alternatives.

- The inclusion of **garment manufacturing** in virgin-based production processes. Producing a garment from polyester yarn is very emissions-intensive relative to raw material and pre-processing (including pelletization) and garment use is also significant. This stage and garment use were both excluded from the comparative analysis in favor of maintaining a comparable basis with bottle-to-fiber and the intended focus of the process on the cradle-to-yarn + EOL steps, as specified by PolyCo. For more detail on the Comparable Basis used for this screening LCA, refer to the section, Comparable Basis, herein.

[2] (Moazzem, et al., 2022): *Inaccurate polyester textile environmental product declarations.* 2022, Volume 2, Issue 1: 47-63. doi: 10.3934/ctr.2022003

- **Bottle-to-fiber comparison** relies on detailed activity data for energy-use by process step drawn from a peer-reviewed study of life cycle energy and GHG emissions of PET recycling. This includes baled PET-to-flake and PET flake-to-pellet ratios, inputs like sodium hydroxide and sulphuric acid, solid waste from processing, and electricity and heat for both PET BTF production and pellet extrusion. For the sake of comparison, both the energy source and EOL treatment parameters used for PolyCo are the same parameter inputs used for BTF. Using different energy sources by technology would result in a range of outputs and results, and would require a thorough sensitivity analysis.

The screening LCA is intended to produce a high-level overview of the major impacts and hotspots of the different phases of the product cradle-to-gate + EOL life cycle and a comparison of where those hotspots differ with related technologies. As the primary goal was to produce an initial estimate of emissions materiality and comparison, sensitivity in factors such as electricity emissions, transportation, and the varying of different partners for process steps including collection & spinning is to be detailed further in a longer full-fledged LCA in Q1 2024.

Overview of the Comparative Materials & Processes

The following table was produced to display inputs, outputs, and any waste streams resulting from the three compared processes, with the PolyCo process explored in further detail than the BTF and virgin processes.

Technology	Virgin Polyester	Bottle-to-Fiber Polyester	Circular Polyester
Inputs	Purified Terephthalic Acid, Ethylene Glycol, Energy, Water*	Baled PET bottles, Sodium hydroxide, Sulphuric acid, Energy, Water*	Baled Feedstock (35% Post-Consumer Waste, 65% Production Overspill), Ethylene Glycol, Potassium Hydroxide, Titanium Oxide, Colorants*, Energy
Outputs	Virgin PET Pellets	Recycled PET Pellets	Circular PET Pellets
Waste Streams	Polyester waste	Polyester waste	Buttons & Zippers (recycled), Ethylene Glycol (recycled), Polyester waste (recycled)

**-to be included in full-fledged LCA in Q1 2024*

Process flow diagrams were created as part of the LCI Inventory process to inform model creation. These diagrams explicate inputs, outputs, and waste streams for the purposes of clearly defining steps to be replicated in modeling formats. The diagrams can be found below in Figures 7.1, 7.2, 7.3, and 7.4.

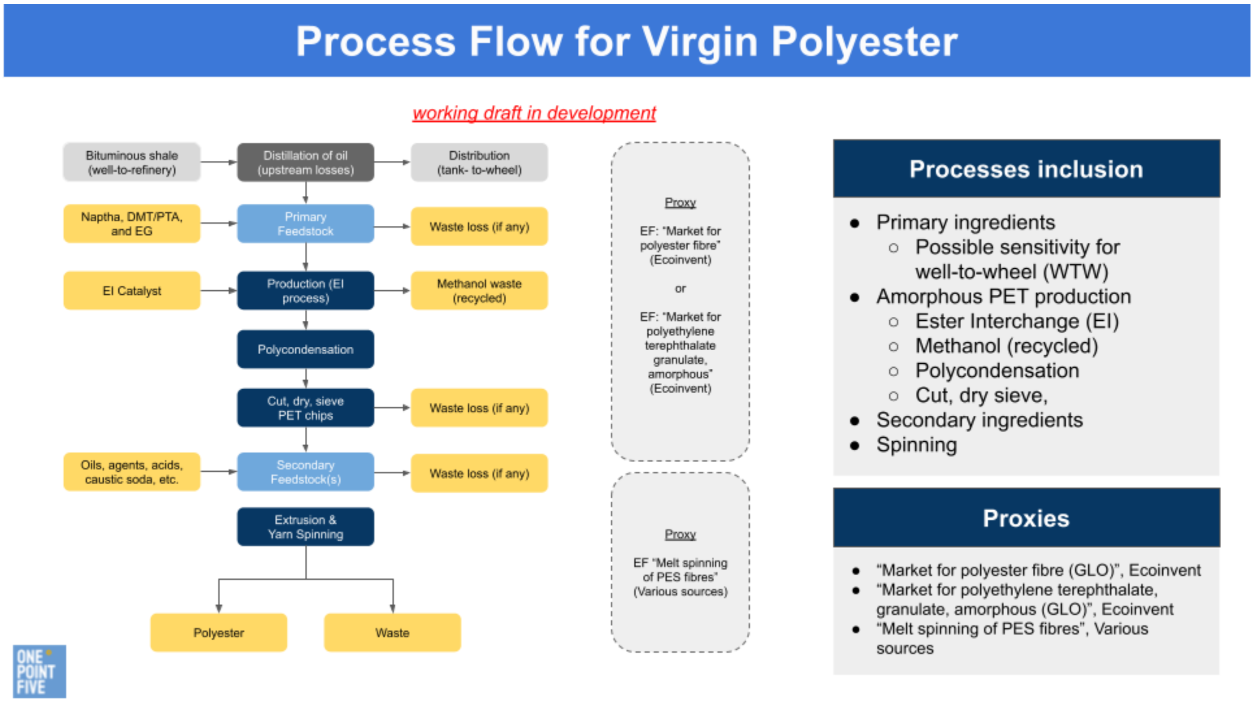


Figure 7.1: LCI Process flow diagram for virgin polyester

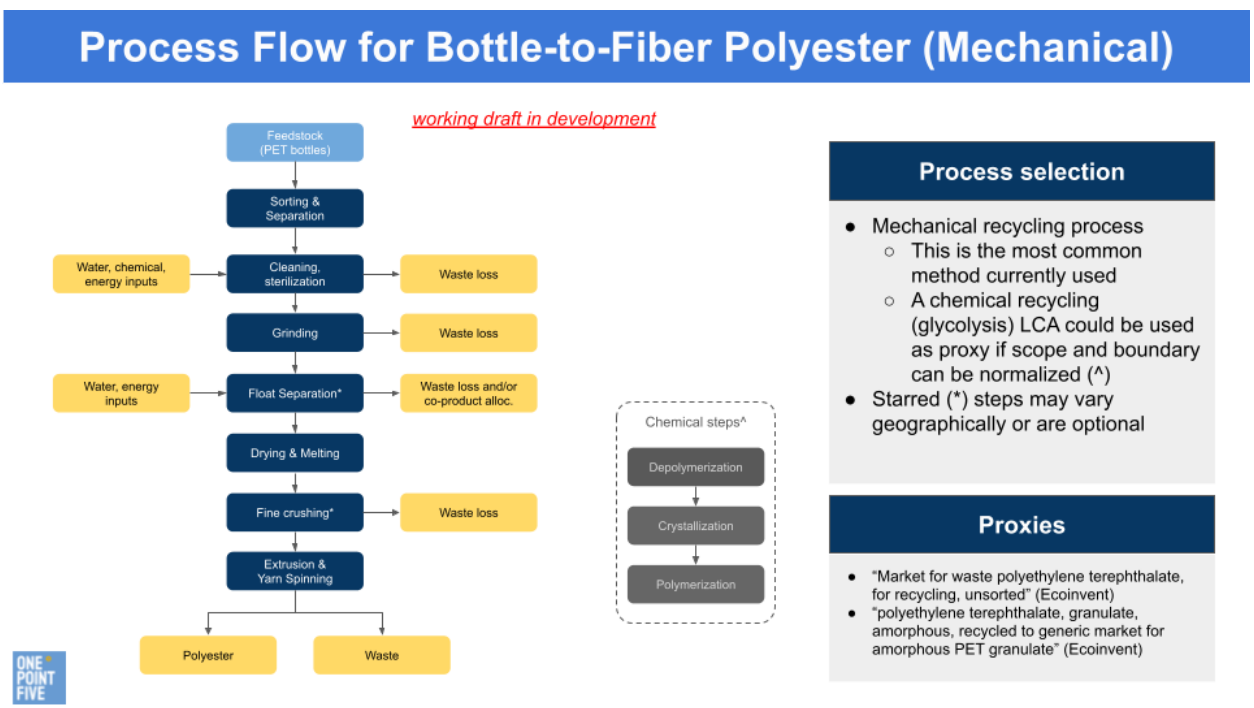


Figure 7.2: LCI Process flow diagram for bottle-to-fiber polyester

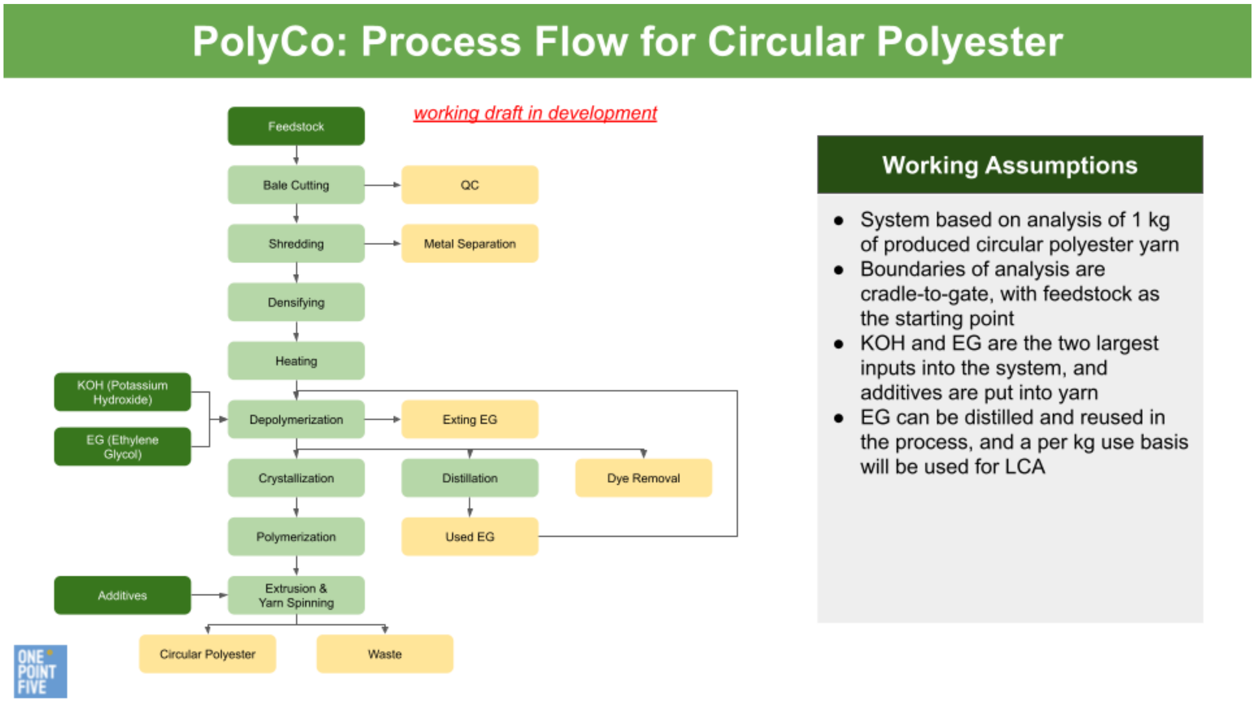


Figure 7.3: LCI Process flow diagram for circular polyester

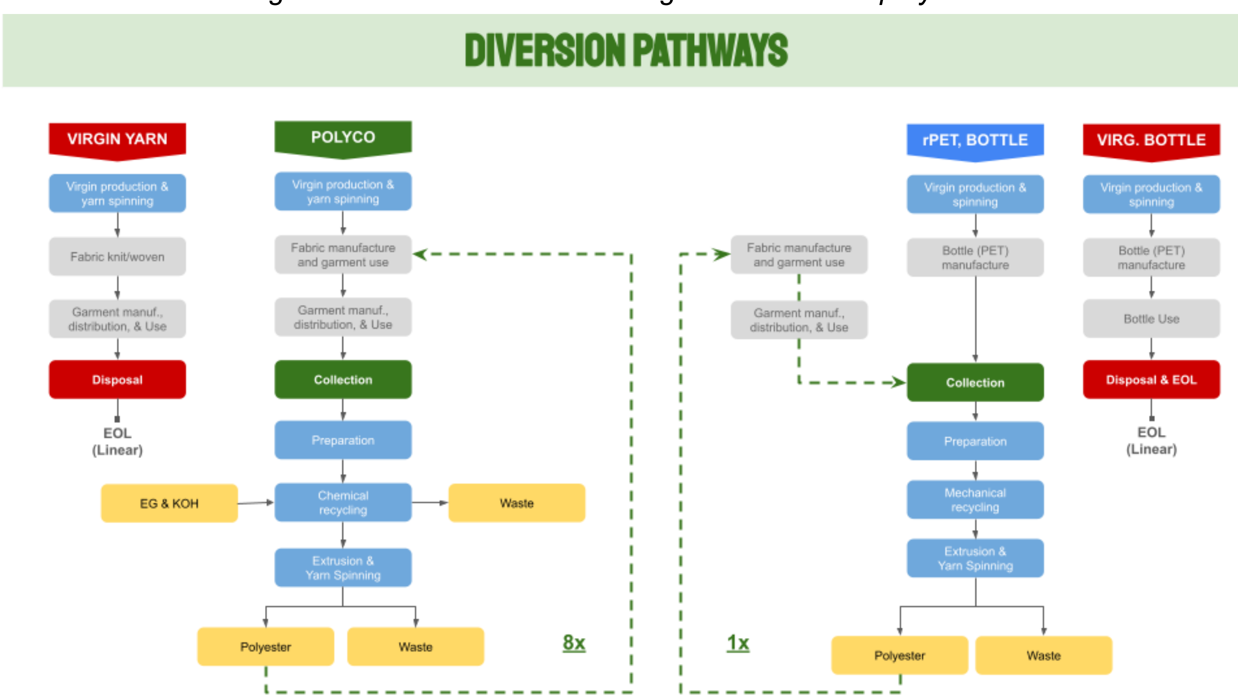


Figure 7.4: LCI Process flow diagram detailing the intersection between the three compared technologies

Additionally, all processes, but for spinning (for PolyCo planning purposes), were assumed to utilize renewable energy.

The PolyCo process utilizes green transport, with the full-fledged LCA to further explore collection and distribution options.

The PolyCo production is based on a prospective facility in Southern Europe, which may eventually expand to include locations in Asia and North America, which will be explored in the full LCA in Q1 2024. All production was assumed to be performed in this prospective location.

Data Sources and Quality

Refer to the "EFs" sheet in the *PolyCo Screening LCA* (Excel file) for a detailed table summary itemizing every emission factor and/or proxy. The regional specificity, source/version (including LCIA method in the case of Ecoinvent secondary emission factors), GWP value, amount, unit of measure (UOM), and a description of the scope and boundary, are included in the mentioned sheet.

From a high-level, the following data sources were relied upon:

- PolyCo:
    - Project Esters case study and data model received from PolyCo including plant feedstock source (waste) input and rPET output, key additives and feedstock ratios or quantities thereof, transportation distances, and energy-intensity.
    - Material flow was reviewed for energy input/output and feedstock, however, the screening models prioritized feedstock activity data from Project Ester based on discussions with client and hotspot analysis of key emission drivers.
    - The emission factor for capital equipment relies on (EPA, 2019) from climatiq.
- Bottle
    - Activity data from peer-reviewed study of the life cycle energy and GHG emissions of PET recycling.[3]
    - Various other secondary emission factors, including from Ecoinvent and other peer reviewed studies, were researched to contextualize the energy and emission factor data from said paper.
    - For a detailed summary of recycled bottle-specific data inputs, refer to the sheet "Bottle" in the *PolyCo Screening LCA* (Excel file).
- Virgin:
    - A cradle-to-cradle life cycle assessment of polyethylene terephthalate by (Tamoor, et al., 2022) was relied on for key feedstock and energy-use amounts for the production of 1 kg virgin PET pellet.[4]

[3] (Moazzem, et al., 2022): *Inaccurate polyester textile environmental product declarations.* 2022, Volume 2, Issue 1: 47-63. doi: 10.3934/ctr.2022003
[4] (Tamoor, et al., 2022): The Cradle-to-Cradle Life Cycle Assessment of Polyethylene terephthalate: Environmental Perpsective. 2022 Feb 28. doi: 10.3390/molecules27051599

- Other secondary emission factors were referenced, including from Ecoinvent for Market for polyethylene terephthalate, granulate, amorphous (GLO), Market for ethylene glycol (GLO), Market for potassium hydroxide (GLO).
- Adjusted emission factors were developed develop proxy emission factors for Adjusted market for ethylene glycol (EU, BAU), Adjusted market for ethylene glycol (EU, renewable), Adjusted PTA (EU, BAU), Adjusted PTA (EU, Renewable), which are key feedstock for virgin PET. This was done in order to perform a sensitivity of the impact of energy sources for virgin PET production to determine the impact on the comparison between virgin PET and PolyCo rPET.
- Other sources were referenced for both virgin PET production and spinning including the *Fiber Bible Part 2* (Sandin, et al., 2019), Polyester fiber production, finished (RoW) from Econinvent, and Mistra Report #5: *Environmental assessment of Swedish clothing consumption* (Sandin, et al., 2019).

- Common data:
  - Spinning relies on sources outlined previously.[5] Various other sources were reviewed and can be referenced in the sheet "**EFs**" and the sheet "**Spinning**" in the *PolyCo Screening LCA* (Excel file).
  - EOL treatment emission factors rely on a selection from Ecovinent with an emphasis on using European regional emission factors where available.
  - Electricity emission factors were researched for various regions and/or countries, however, the final datasets being used in the model and which can be used live for a sensitivity analysis is recent, average LCA GHG-intensity of electric energy sources from (NREL, 2021).[6]
  - For waste collection, bale loading emission factors were proxied using various data points; this process is immaterial for the purpose of the comparison herein.
  - For transportation and distribution by cargo or HGV (diesel), and plastic packaging associated with bales, emission factors rely on GHG reporting conversion factors from (UK BEIS, 2023).[7]

Sensitivity

The *PolyCo Screening LCA* (Excel file) provides multiple key parameter assumption drop-downs summarized in the sheet, "Key_Scenario_drivers" and available as drop-downs in the sheet "Working Model". They include: (1) Virgin feedstock, (2) Spinning energy-intensity, (3) Spinning energy-source scenario, (4) Feedstock source for PolyCo (% post-consuming versus % spill), (5) Energy-source for PolyCo and Bottle manufacturing, (6) EOL treatment. Each of these parameter inputs can be used to generate a different LCA screening sensitivity or hotspot analysis. This can be studied using the sheet, "Summary", which presents the key parameters at the top and shows the resulting GHG emissions for Virgin, PolyCo, and Bottle by (GLO, BAU),

[5] (Moazzem, et al., 2022): *Inaccurate polyester textile environmental product declarations*. 2022, Volume 2, Issue 1: 47-63. doi: 10.3934/ctr.2022003
[6] (NREL, 2021): Life Cycle Greenhouse Gas Emissions from Electricity Generation: Update (nrel.gov)
[7] (BEIS, 2023): https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2023

(EU, BAU), and (EU, Renewable). The parameter assumptions available via drop-down in the sheet, “Working Model” are as follows:

1. Virgin feedstock has three scenarios: (GLO, BAU), (EU, BAU), and (EU, Renewable), which affects the energy-source and emission-intensity of virgin PET pellet production.
2. Spinning energy-intensity has three scenarios: Min, Max, and Average energy-intensity per unit PET pellet.
3. Spinning energy-source has the same scenarios as Energy-source for PolyCo.
4. Feedstock source (% post-consumer v. spill) for PolyCo determines the % feedstock source from post-consumer waste versus spill. The model updates the associated material inputs, loss factors, etc.
5. Energy-source for PolyCo has elevent scenario sources from (NERL, 2021) LCA emission factor for commercial electricity-generating technologies: biomass, photovoltaic, concentrating solar power, geothermal, hydropower, ocean, wind, nuclear, natural gas, oil, and coal.
6. EOL treatment has five available treatments for PET waste from Ecoinvent: 1% open dump and 99% incineration; 100% open burning; 100% open dump; 100% unsanitary landfill; and a hypothetical sensitivity emission factor of zero (0).

The *PolyCo Screening LCA* (Excel file) and figures which are presented within this paper rely on the key parameter assumptions as follows (summarized in the sheet, “Key_Scenario_drivers”).

1. Virgin feedstock has three scenarios: (EU, Renewable)
2. Spinning energy-intensity: Average energy-intensity
3. Spinning energy-source: Oil
4. Feedstock source (% post-consumer v. spill): 35/65%
5. Energy-source for PolyCo: photovoltaic
6. EOL treatment: 1% open dump/ 99% incineration (CH)

Study Fundamentals

This screening was performed by OnePointFive, a climate advisory group headquartered in New York City, on behalf of PolyCo, operating in North Carolina, and managed in Sweden. The model & results have yet to be verified by a third party and will be followed by & expanded upon by a full LCA in Q1 2024.